Лабораторная работа №1

Выполнила: Малей Ангелина, гр. 453503. Вариант 19.

Тема: Система контроля версий Git.

Цель: Познакомиться с возможностями и получить практические навыки работы с системой контроля версий Git.

Задание:

1 Подготовка рабочего окружения (ОС Windows)

a) Установка Git

```
C:\Users\Angelina> git --version
git version 2.53.0.windows.1
```

b) Выполнение базовых настроек

```
Angelina@angelpc MINGW64 ~
$ git config --global user.name "Angelina Malei"

Angelina@angelpc MINGW64 ~
$ git config --global user.email "maley.gelya@mail.ru"
```

```
Angelina@angelpc MINGW64 ~
$ git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=schannel
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=main
user.name=Angelina Malei
user.email=maley.gelya@mail.ru
core.editor=notepad.exe
```

2 Создание структуры папок с помощью командной строки и их размещение в локальном репозитории Git на своем компьютере

a) Создание основной структуры папок:

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ
$ mkdir 453503_MALEY_19 && cd 453503_MALEY_19

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19
$ mkdir IGI && cd IGI

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19/IGI
$ mkdir LR1 LR2 LR3 LR4 LR5

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19/IGI
$ dir
LR1  LR2  LR3  LR4  LR5

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19/IGI
$ cd .. && mkdir STRWEB && cd STRWEB

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19/STRWEB
$ mkdir LR1 LR2 LR3 LR4

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19/STRWEB
$ dir
LR1  LR2  LR3  LR4
```

b) Создание структуры согласно индивидуальному заданию:

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19
$ cd IGI/LR1

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19/IGI/LR1
$ mkdir bin && cd bin && mkdir 676 && mkdir rm && cd rm && mkdir 676

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19/IGI/LR1/bin/rm
$ cd ../../ && mkdir bar && cd bar && mkdir 766 && mkdir baz

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19/IGI/LR1/bar
$ cd baz && mkdir 766 && mkdir example && cd example && mkdir 677 && cd ..

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19/IGI/LR1/bar/baz
$ echo 444 > readme.txt

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19/IGI/LR1/bar/baz
$ cd ../../ && mkdir foo && cd foo && mkdir 111

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19/IGI/LR1/foo
$ echo 777 > test.txt
```

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19/IGI/LR1
$ tree
.
|-- bar
|   |-- 766
|   `-- baz
|       |-- 766
|       |-- example
|       |   `-- 677
|       `-- readme.txt
|-- bin
|   |-- 676
|   `-- rm
|       `-- 676
`-- foo
    |-- 111
    `-- test.txt

12 directories, 2 files
```

c) Создание в каждой папке текстового файла в соответствии с ее названием:

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19
$ touch 453503_MALEY_19Test.txt

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19
$ touch IGI/IGITest.txt && touch STRWEB/STRWEBTest.txt
```

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19/IGI
$ touch LR1/LR1Test.txt && touch LR2/LR2Test.txt && touch LR3/LR3Test.txt && touch LR4/LR4Test.txt && touch LR5/LR5Test.txt

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19/IGI
$ cd .. && cd STRWEB && touch LR1/LR1Test.txt && touch LR2/LR2Test.txt && touch LR3/LR3Test.txt && touch LR4/LR4Test.txt
```

Результат:

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19
$ tree
.
|-- 453503_MALEY_19Test.txt
|-- IGI
|   |-- IGITest.txt
|   |-- LR1
|   |   |-- LR1Test.txt
|   |   |-- bar
|   |   |   |-- 766
|   |   |   `-- baz
|   |   |       |-- 766
|   |   |       |-- example
|   |   |       |   `-- 677
|   |   |       `-- readme.txt
|   |   |-- bin
|   |   |   |-- 676
|   |   |   `-- rm
|   |   |       `-- 676
|   |   `-- foo
|   |       |-- 111
|   |       `-- test.txt
|   |-- LR2
|   |   `-- LR2Test.txt
|   |-- LR3
|   |   `-- LR3Test.txt
|   |-- LR4
|   |   `-- LR4Test.txt
|   `-- LR5
|       `-- LR5Test.txt
`-- STRWEB
    |-- LR1
    |   `-- LR1Test.txt
    |-- LR2
    |   `-- LR2Test.txt
    |-- LR3
    |   `-- LR3Test.txt
    |-- LR4
    |   `-- LR4Test.txt
    `-- STRWEBTest.txt
```

d) Создание репозитория:

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19
$ git status
fatal: not a git repository (or any of the parent directories): .git

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19
$ git init
Initialized empty Git repository in C:/Users/Angelina/Desktop/4sem/ИГИ/453503_MALEY_19/.git/

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        453503_MALEY_19Test.txt
        IGI/
        STRWEB/

nothing added to commit but untracked files present (use "git add" to track)
```

3 Создание репозитория внутри папки LR1 с помощью клонирования проекта с сайта github.com *(https://github.com/smartiqaorg/geometric_lib)*

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ cd IGI/LR1

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19/IGI/LR1 (main)
$ git clone https://github.com/smartiqaorg/geometric_lib.git
Cloning into 'geometric_lib'...
remote: Enumerating objects: 35, done.
remote: Total 35 (delta 0), reused 0 (delta 0), pack-reused 35 (from 1)
Receiving objects: 100% (35/35), 4.69 KiB | 320.00 KiB/s, done.
Resolving deltas: 100% (5/5), done.
```

4 Выполнение следующих действий, каждый раз проверяя состояние файлов в локальном репозитории до и после выполнения операции:

a) Сделать индексацию нескольких файлов в вашем Git-репозитории (git add ...)

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19/IGI/LR1/geometric_lib (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19/IGI/LR1/geometric_lib (main)
$ git add circle.py square.py

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19/IGI/LR1/geometric_lib (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
```

b) Внести изменения в файлы

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19/IGI/LR1/geometric_lib (main)
$ echo "# Hello, my name is Angelina!" >> square.py

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19/IGI/LR1/geometric_lib (main)
$ cat square.py

def area(a):
    return a * a


def perimeter(a):
    return 4 * a
# Hello, my name is Angelina!

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19/IGI/LR1/geometric_lib (main)
$ git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   square.py

no changes added to commit (use "git add" and/or "git commit -a")
```

c) Выполнить коммиты к проиндексированным файлам

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19/IGI/LR1/geometric_lib (main)
$ git add square.py
warning: in the working copy of 'square.py', LF will be replaced by CRLF the next time Git

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19/IGI/LR1/geometric_lib (main)
$ git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   square.py


Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19/IGI/LR1/geometric_lib (main)
$ git commit -m "Добавлена строка в файл square.py"
[main c662ff0] Добавлена строка в файл square.py
 1 file changed, 1 insertion(+)

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19/IGI/LR1/geometric_lib (main)
$ git status
On branch main
Your branch is ahead of 'origin/main' by 2 commits.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
```

d) Добавить и удалить файлы (один и группу файлов с определенным расширением)

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19/IGI/LR1/geometric_lib (main)
$ touch primer.py && touch test.txt && touch proverka.txt

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19/IGI/LR1/geometric_lib (main)
$ git status
On branch main
Your branch is ahead of 'origin/main' by 2 commits.
  (use "git push" to publish your local commits)

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        primer.py
        proverka.txt
        test.txt

nothing added to commit but untracked files present (use "git add" to track)
```

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19/IGI/LR1/geometric_lib (main)
$ git add primer.py proverka.txt test.txt

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19/IGI/LR1/geometric_lib (main)
$ git status
On branch main
Your branch is ahead of 'origin/main' by 2 commits.
  (use "git push" to publish your local commits)

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   primer.py
        new file:   proverka.txt
        new file:   test.txt
```

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19/IGI/LR1/geometric_lib (main)
$ git rm -rf primer.py
rm 'primer.py'

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19/IGI/LR1/geometric_lib (main)
$ git status
On branch main
Your branch is ahead of 'origin/main' by 2 commits.
  (use "git push" to publish your local commits)

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   proverka.txt
        new file:   test.txt


Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19/IGI/LR1/geometric_lib (main)
$ git rm -rf *.txt
rm 'proverka.txt'
rm 'test.txt'

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19/IGI/LR1/geometric_lib (main)
$ git status
On branch main
Your branch is ahead of 'origin/main' by 2 commits.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
```

e) Переименовать файлы

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19/IGI/LR1/geometric_lib (main)
$ git mv circle.py kryg.py

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19/IGI/LR1/geometric_lib (main)
$ git mv square.py kvadrat.py

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19/IGI/LR1/geometric_lib (main)
$ git status
On branch main
Your branch is ahead of 'origin/main' by 2 commits.
  (use "git push" to publish your local commits)

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        renamed:    circle.py -> kryg.py
        renamed:    square.py -> kvadrat.py
```

f) Создать для IGI и STRWEB, а также для каждой ЛР, вложенной в них, <u>отдельные ветки</u>, переместить папки и файлы на соответствующие ветки

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git checkout -b IGI
Switched to a new branch 'IGI'

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (IGI)
$ git status
On branch IGI
nothing to commit, working tree clean

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (IGI)
$ rm -rf STRWEB

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (IGI)
$ rm -f 453503_MALEY_19Test.txt

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (IGI)
$ git status
On branch IGI
Changes not staged for commit:
  (use "git add/rm <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        deleted:    453503_MALEY_19Test.txt
        deleted:    STRWEB/LR1/LR1Test.txt
        deleted:    STRWEB/LR2/LR2Test.txt
        deleted:    STRWEB/LR3/LR3Test.txt
        deleted:    STRWEB/LR4/LR4Test.txt
        deleted:    STRWEB/STRWEBTest.txt

no changes added to commit (use "git add" and/or "git commit -a")

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (IGI)
$ git add .

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (IGI)
$ git status
On branch IGI
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        deleted:    453503_MALEY_19Test.txt
        deleted:    STRWEB/LR1/LR1Test.txt
        deleted:    STRWEB/LR2/LR2Test.txt
        deleted:    STRWEB/LR3/LR3Test.txt
        deleted:    STRWEB/LR4/LR4Test.txt
        deleted:    STRWEB/STRWEBTest.txt


Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (IGI)
$ git commit -m "created IGI branch"
[IGI 1c84e30] created IGI branch
 6 files changed, 1 deletion(-)
 delete mode 100644 453503_MALEY_19Test.txt
 delete mode 100644 STRWEB/LR1/LR1Test.txt
 delete mode 100644 STRWEB/LR2/LR2Test.txt
 delete mode 100644 STRWEB/LR3/LR3Test.txt
 delete mode 100644 STRWEB/LR4/LR4Test.txt
 delete mode 100644 STRWEB/STRWEBTest.txt
```

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (IGI)
$ git checkout main
Switched to branch 'main'

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git status
On branch main
nothing to commit, working tree clean
```

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git checkout -b STRWEB
Switched to a new branch 'STRWEB'

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (STRWEB)
$ git status
On branch STRWEB
nothing to commit, working tree clean

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (STRWEB)
$ rm -rf IGI && rm -f 453503_MALEY_19Test.txt

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (STRWEB)
$ git status
On branch STRWEB
Changes not staged for commit:
  (use "git add/rm <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        deleted:    453503_MALEY_19Test.txt
        deleted:    IGI/IGITest.txt
        deleted:    IGI/LR1/LR1Test.txt
        deleted:    IGI/LR1/bar/baz/readme.txt
        deleted:    IGI/LR1/foo/test.txt
        deleted:    IGI/LR2/LR2Test.txt
        deleted:    IGI/LR3/LR3Test.txt
        deleted:    IGI/LR4/LR4Test.txt
        deleted:    IGI/LR5/LR5Test.txt

no changes added to commit (use "git add" and/or "git commit -a")

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (STRWEB)
$ git add .

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (STRWEB)
$ git commit -m "created STRWEB branch"
[STRWEB e76c10b] created STRWEB branch
 9 files changed, 3 deletions(-)
 delete mode 100644 453503_MALEY_19Test.txt
 delete mode 100644 IGI/IGITest.txt
 delete mode 100644 IGI/LR1/LR1Test.txt
 delete mode 100644 IGI/LR1/bar/baz/readme.txt
 delete mode 100644 IGI/LR1/foo/test.txt
 delete mode 100644 IGI/LR2/LR2Test.txt
 delete mode 100644 IGI/LR3/LR3Test.txt
 delete mode 100644 IGI/LR4/LR4Test.txt
 delete mode 100644 IGI/LR5/LR5Test.txt

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (STRWEB)
$ git status
On branch STRWEB
nothing to commit, working tree clean
```

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (STRWEB)
$ git status
On branch STRWEB
nothing to commit, working tree clean

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (STRWEB)
$ git checkout -b STRWEB_LR1
Switched to a new branch 'STRWEB_LR1'

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (STRWEB_LR1)
$ git status
On branch STRWEB_LR1
nothing to commit, working tree clean

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (STRWEB_LR1)
$ rm -rf STRWEB/LR2 STRWEB/LR3 STRWEB/LR4

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (STRWEB_LR1)
$ git status
On branch STRWEB_LR1
Changes not staged for commit:
  (use "git add/rm <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        deleted:    STRWEB/LR2/LR2Test.txt
        deleted:    STRWEB/LR3/LR3Test.txt
        deleted:    STRWEB/LR4/LR4Test.txt

no changes added to commit (use "git add" and/or "git commit -a")

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (STRWEB_LR1)
$ rm -f STRWEB/STRWEBTest.txt

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (STRWEB_LR1)
$ git status
On branch STRWEB_LR1
Changes not staged for commit:
  (use "git add/rm <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        deleted:    STRWEB/LR2/LR2Test.txt
        deleted:    STRWEB/LR3/LR3Test.txt
        deleted:    STRWEB/LR4/LR4Test.txt
        deleted:    STRWEB/STRWEBTest.txt

no changes added to commit (use "git add" and/or "git commit -a")

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (STRWEB_LR1)
$ git add .

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (STRWEB_LR1)
$ git commit -m "created STRWEB_LR1 branch"
[STRWEB_LR1 a7d2444] created STRWEB_LR1 branch
 4 files changed, 0 insertions(+), 0 deletions(-)
 delete mode 100644 STRWEB/LR2/LR2Test.txt
 delete mode 100644 STRWEB/LR3/LR3Test.txt
 delete mode 100644 STRWEB/LR4/LR4Test.txt
 delete mode 100644 STRWEB/STRWEBTest.txt

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (STRWEB_LR1)
$ git status
On branch STRWEB_LR1
nothing to commit, working tree clean
```

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (STRWEB_LR1)
$ git status
On branch STRWEB_LR1
nothing to commit, working tree clean

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (STRWEB_LR1)
$ git checkout STRWEB
Switched to branch 'STRWEB'

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (STRWEB)
$ git status
On branch STRWEB
nothing to commit, working tree clean
```

По аналогии создаем ветки STRWEB_LR2, STRWEB_LR3, STRWEB_LR4. После этого переходим на ветку IGI и создаем ветки там.

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (STRWEB_LR4)
$ git status
On branch STRWEB_LR4
nothing to commit, working tree clean

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (STRWEB_LR4)
$ git checkout IGI
Switched to branch 'IGI'

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (IGI)
$ git status
On branch IGI
nothing to commit, working tree clean
```

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (IGI)
$ git checkout -b IGI_LR1
Switched to a new branch 'IGI_LR1'

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (IGI_LR1)
$ git status
On branch IGI_LR1
nothing to commit, working tree clean

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (IGI_LR1)
$ rm -rf IGI/LR2 IGI/LR3 IGI/LR4 IGI/LR5 && rm -f IGI/IGITest.txt

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (IGI_LR1)
$ git status
On branch IGI_LR1
Changes not staged for commit:
  (use "git add/rm <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        deleted:    IGI/IGITest.txt
        deleted:    IGI/LR2/LR2Test.txt
        deleted:    IGI/LR3/LR3Test.txt
        deleted:    IGI/LR4/LR4Test.txt
        deleted:    IGI/LR5/LR5Test.txt

no changes added to commit (use "git add" and/or "git commit -a")

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (IGI_LR1)
$ git add .

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (IGI_LR1)
$ git status
On branch IGI_LR1
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        deleted:    IGI/IGITest.txt
        deleted:    IGI/LR2/LR2Test.txt
        deleted:    IGI/LR3/LR3Test.txt
        deleted:    IGI/LR4/LR4Test.txt
        deleted:    IGI/LR5/LR5Test.txt


Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (IGI_LR1)
$ git commit -m "created IGI_LR1 branch"
[IGI_LR1 70691f8] created IGI_LR1 branch
 5 files changed, 0 insertions(+), 0 deletions(-)
 delete mode 100644 IGI/IGITest.txt
 delete mode 100644 IGI/LR2/LR2Test.txt
 delete mode 100644 IGI/LR3/LR3Test.txt
 delete mode 100644 IGI/LR4/LR4Test.txt
 delete mode 100644 IGI/LR5/LR5Test.txt

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (IGI_LR1)
$ git status
On branch IGI_LR1
nothing to commit, working tree clean
```

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (IGI_LR1)
$ git checkout IGI
Switched to branch 'IGI'

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (IGI)
$ git status
On branch IGI
nothing to commit, working tree clean
```

По аналогии создаем ветки IGI_LR2, IGI_LR3, IGI_LR4, IGI_LR5.

Результат:

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git graph
* created IGI_LR5 branch  (IGI_LR5)
| * created IGI_LR4 branch  (IGI_LR4)
|/
| * created IGI_LR3 branch  (IGI_LR3)
|/
| * created IGI_LR2 branch  (IGI_LR2)
|/
| * created IGI_LR1 branch  (IGI_LR1)
|/
* created IGI branch  (IGI)
| * created STRWEB_LR4 branch  (STRWEB_LR4)
| | * created SRTWEB_LR3 branch  (STRWEB_LR3)
| |/
| | * created STRWEB_LR2 branch  (STRWEB_LR2)
| |/
| | * created STRWEB_LR1 branch  (STRWEB_LR1)
| |/
| * created STRWEB branch  (STRWEB)
|/
* committed all .txt files  (HEAD -> main)
* added .gitignore into main
* initialized empty repo
```

g) Вывести историю коммитов: полностью и с ограничением вывода (флаги n, since, after, until, before, author, committer, grep)

<u>Вывод полной истории коммитов:</u>

Вывод истории коммитов – команда **git log**

Флаги: **--all** – все коммиты

--oneline – сокращенный вывод

Все коммиты на текущей ветке:

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git log
commit 77deff2ba33ca3965a75f05fee3d312c055e350d (HEAD -> main)
Author: Angelina Malei <maley.gelya@mail.ru>
Date:   Sun Feb 22 18:57:25 2026 +0300

    committed all .txt files

commit 4cbe11f6e8437f6149e8dadcb0f78a5119dcd61d
Author: Angelina Malei <maley.gelya@mail.ru>
Date:   Sun Feb 22 18:52:15 2026 +0300

    added .gitignore into main

commit 304cb7de043853e80ba659f91d0473509e9a47f2
Author: Angelina Malei <maley.gelya@mail.ru>
Date:   Sun Feb 22 18:33:31 2026 +0300

    initialized empty repo
```

Коммиты со всех веток:

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git log --all
commit 3c10766f5719950660f013558ee5436c97f52b6d (IGI_LR5)
Author: Angelina Malei <maley.gelya@mail.ru>
Date:   Sun Feb 22 19:49:33 2026 +0300

    created IGI_LR5 branch

commit eb7d5106cb4d9ae26d4671f01877d50ff5255232 (IGI_LR4)
Author: Angelina Malei <maley.gelya@mail.ru>
Date:   Sun Feb 22 19:47:27 2026 +0300

    created IGI_LR4 branch
```

Сокращенный вывод коммитов на текущей ветке:

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git log --oneline
77deff2 (HEAD -> main) committed all .txt files
4cbe11f added .gitignore into main
304cb7d initialized empty repo
```

Вывод истории коммитов с ограничениями:

Последние три коммита (флаг **-n**):

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git log --all -3
commit 3c10766f5719950660f013558ee5436c97f52b6d (IGI_LR5)
Author: Angelina Malei <maley.gelya@mail.ru>
Date:   Sun Feb 22 19:49:33 2026 +0300

    created IGI_LR5 branch

commit eb7d5106cb4d9ae26d4671f01877d50ff5255232 (IGI_LR4)
Author: Angelina Malei <maley.gelya@mail.ru>
Date:   Sun Feb 22 19:47:27 2026 +0300

    created IGI_LR4 branch

commit 11d6ac0f92ad4e8fe0c8569b357cb3db42c79b3d (IGI_LR3)
Author: Angelina Malei <maley.gelya@mail.ru>
Date:   Sun Feb 22 19:45:15 2026 +0300

    created IGI_LR3 branch
```

Коммиты после определенной даты (флаги **--since/--after**):

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git log --oneline --all --since="2026-02-14"
3c10766 (IGI_LR5) created IGI_LR5 branch
eb7d510 (IGI_LR4) created IGI_LR4 branch
11d6ac0 (IGI_LR3) created IGI_LR3 branch
17be4b6 (IGI_LR2) created IGI_LR2 branch
70691f8 (IGI_LR1) created IGI_LR1 branch
971c9b5 (STRWEB_LR4) created STRWEB_LR4 branch
b5ab026 (STRWEB_LR3) created SRTWEB_LR3 branch
1de7e07 (STRWEB_LR2) created STRWEB_LR2 branch
a7d2444 (STRWEB_LR1) created STRWEB_LR1 branch
e76c10b (STRWEB) created STRWEB branch
1c84e30 (IGI) created IGI branch
77deff2 (HEAD -> main) committed all .txt files
4cbe11f added .gitignore into main
304cb7d initialized empty repo

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git log --oneline --all --after="2026-02-02"
3c10766 (IGI_LR5) created IGI_LR5 branch
eb7d510 (IGI_LR4) created IGI_LR4 branch
11d6ac0 (IGI_LR3) created IGI_LR3 branch
17be4b6 (IGI_LR2) created IGI_LR2 branch
70691f8 (IGI_LR1) created IGI_LR1 branch
971c9b5 (STRWEB_LR4) created STRWEB_LR4 branch
b5ab026 (STRWEB_LR3) created SRTWEB_LR3 branch
1de7e07 (STRWEB_LR2) created STRWEB_LR2 branch
a7d2444 (STRWEB_LR1) created STRWEB_LR1 branch
e76c10b (STRWEB) created STRWEB branch
1c84e30 (IGI) created IGI branch
77deff2 (HEAD -> main) committed all .txt files
4cbe11f added .gitignore into main
304cb7d initialized empty repo
```

Коммиты до определенной даты (флаги **--until/--before**):

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git log --oneline --all --before="2026-02-02"

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git log --oneline --all --before="2026-02-23"
3c10766 (IGI_LR5) created IGI_LR5 branch
eb7d510 (IGI_LR4) created IGI_LR4 branch
11d6ac0 (IGI_LR3) created IGI_LR3 branch
17be4b6 (IGI_LR2) created IGI_LR2 branch
70691f8 (IGI_LR1) created IGI_LR1 branch
971c9b5 (STRWEB_LR4) created STRWEB_LR4 branch
b5ab026 (STRWEB_LR3) created SRTWEB_LR3 branch
1de7e07 (STRWEB_LR2) created STRWEB_LR2 branch
a7d2444 (STRWEB_LR1) created STRWEB_LR1 branch
e76c10b (STRWEB) created STRWEB branch
1c84e30 (IGI) created IGI branch
77deff2 (HEAD -> main) committed all .txt files
4cbe11f added .gitignore into main
304cb7d initialized empty repo

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git log --oneline --all --until="2026-02-23"
3c10766 (IGI_LR5) created IGI_LR5 branch
eb7d510 (IGI_LR4) created IGI_LR4 branch
11d6ac0 (IGI_LR3) created IGI_LR3 branch
17be4b6 (IGI_LR2) created IGI_LR2 branch
70691f8 (IGI_LR1) created IGI_LR1 branch
971c9b5 (STRWEB_LR4) created STRWEB_LR4 branch
b5ab026 (STRWEB_LR3) created SRTWEB_LR3 branch
1de7e07 (STRWEB_LR2) created STRWEB_LR2 branch
a7d2444 (STRWEB_LR1) created STRWEB_LR1 branch
e76c10b (STRWEB) created STRWEB branch
1c84e30 (IGI) created IGI branch
77deff2 (HEAD -> main) committed all .txt files
4cbe11f added .gitignore into main
304cb7d initialized empty repo
```

Коммиты по автору файлов (флаг **--author**):

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git log --oneline --author="Angelina Malei"
77deff2 (HEAD -> main) committed all .txt files
4cbe11f added .gitignore into main
304cb7d initialized empty repo
```

Коммиты по автору коммитов (флаг **--committer**):

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git log --oneline --all --committer="Angelina Malei"
3c10766 (IGI_LR5) created IGI_LR5 branch
eb7d510 (IGI_LR4) created IGI_LR4 branch
11d6ac0 (IGI_LR3) created IGI_LR3 branch
17be4b6 (IGI_LR2) created IGI_LR2 branch
70691f8 (IGI_LR1) created IGI_LR1 branch
971c9b5 (STRWEB_LR4) created STRWEB_LR4 branch
b5ab026 (STRWEB_LR3) created SRTWEB_LR3 branch
1de7e07 (STRWEB_LR2) created STRWEB_LR2 branch
a7d2444 (STRWEB_LR1) created STRWEB_LR1 branch
e76c10b (STRWEB) created STRWEB branch
1c84e30 (IGI) created IGI branch
77deff2 (HEAD -> main) committed all .txt files
4cbe11f added .gitignore into main
304cb7d initialized empty repo
```

Коммиты, содержащие <слово> (флаг **--grep**):

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git log --oneline --all --grep="add"
4cbe11f added .gitignore into main
```

h) Исследовать флаги для git log (-p, --stat, --shortstat, --name-only, --name-status, --relative-date, --pretty, --graph):

Полные изменения для каждого коммита (флаг **-p**):

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git log -p
```

```
commit 4cbe11f6e8437f6149e8dadcb0f78a5119dcd61d
Author: Angelina Malei <maley.gelya@mail.ru>
Date:   Sun Feb 22 18:52:15 2026 +0300

    added .gitignore into main

diff --git a/.gitignore b/.gitignore
new file mode 100644
index 0000000..3fdb35e
--- /dev/null
+++ b/.gitignore
@@ -0,0 +1 @@
+IGI/LR1/geometric_lib/
```

Статистика изменений по файлам (флаг **--stat**):

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git log --stat
commit 77deff2ba33ca3965a75f05fee3d312c055e350d (HEAD -> main)
Author: Angelina Malei <maley.gelya@mail.ru>
Date:   Sun Feb 22 18:57:25 2026 +0300

    committed all .txt files

 IGI/IGITest.txt             | 0
 IGI/LR1/LR1Test.txt         | 0
 IGI/LR1/bar/baz/readme.txt  | 1 +
 IGI/LR1/foo/test.txt        | 1 +
 IGI/LR2/LR2Test.txt         | 0
 IGI/LR3/LR3Test.txt         | 0
 IGI/LR4/LR4Test.txt         | 0
 IGI/LR5/LR5Test.txt         | 0
 STRWEB/LR1/LR1Test.txt      | 0
 STRWEB/LR2/LR2Test.txt      | 0
 STRWEB/LR3/LR3Test.txt      | 0
 STRWEB/LR4/LR4Test.txt      | 0
 STRWEB/STRWEBTest.txt       | 0
 13 files changed, 2 insertions(+)
```

Краткая статистика – только итоговая строка (флаг **--shortstat**):

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git log --shortstat
commit 77deff2ba33ca3965a75f05fee3d312c055e350d (HEAD -> main)
Author: Angelina Malei <maley.gelya@mail.ru>
Date:   Sun Feb 22 18:57:25 2026 +0300

    committed all .txt files

 13 files changed, 2 insertions(+)
```

Показывает только имена измененных файлов (флаг **--name-only**):

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git log --name-only
commit 77deff2ba33ca3965a75f05fee3d312c055e350d (HEAD -> main)
Author: Angelina Malei <maley.gelya@mail.ru>
Date:   Sun Feb 22 18:57:25 2026 +0300

    committed all .txt files

IGI/IGITest.txt
IGI/LR1/LR1Test.txt
IGI/LR1/bar/baz/readme.txt
IGI/LR1/foo/test.txt
IGI/LR2/LR2Test.txt
IGI/LR3/LR3Test.txt
IGI/LR4/LR4Test.txt
IGI/LR5/LR5Test.txt
STRWEB/LR1/LR1Test.txt
STRWEB/LR2/LR2Test.txt
STRWEB/LR3/LR3Test.txt
STRWEB/LR4/LR4Test.txt
STRWEB/STRWEBTest.txt
```

Показывает имена файлов и статус изменений (флаг **--name-status**):
Статусы: A (Added), M (Modified), D (Deleted), R (Renamed)

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git log --name-status
commit 77deff2ba33ca3965a75f05fee3d312c055e350d (HEAD -> main)
Author: Angelina Malei <maley.gelya@mail.ru>
Date:   Sun Feb 22 18:57:25 2026 +0300

    committed all .txt files

A       IGI/IGITest.txt
A       IGI/LR1/LR1Test.txt
A       IGI/LR1/bar/baz/readme.txt
A       IGI/LR1/foo/test.txt
A       IGI/LR2/LR2Test.txt
A       IGI/LR3/LR3Test.txt
A       IGI/LR4/LR4Test.txt
A       IGI/LR5/LR5Test.txt
A       STRWEB/LR1/LR1Test.txt
A       STRWEB/LR2/LR2Test.txt
A       STRWEB/LR3/LR3Test.txt
A       STRWEB/LR4/LR4Test.txt
A       STRWEB/STRWEBTest.txt
```

Рисует граф истории коммитов с ветками (флаг **--graph**):

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git log --graph --all --oneline
* 3c10766 (IGI_LR5) created IGI_LR5 branch
| * eb7d510 (IGI_LR4) created IGI_LR4 branch
|/
| * 11d6ac0 (IGI_LR3) created IGI_LR3 branch
|/
| * 17be4b6 (IGI_LR2) created IGI_LR2 branch
|/
| * 70691f8 (IGI_LR1) created IGI_LR1 branch
|/
* 1c84e30 (IGI) created IGI branch
| * 971c9b5 (STRWEB_LR4) created STRWEB_LR4 branch
| | * b5ab026 (STRWEB_LR3) created SRTWEB_LR3 branch
| |/
| | * 1de7e07 (STRWEB_LR2) created STRWEB_LR2 branch
| |/
| | * a7d2444 (STRWEB_LR1) created STRWEB_LR1 branch
| |/
| * e76c10b (STRWEB) created STRWEB branch
|/
* 77deff2 (HEAD -> main) committed all .txt files
* 4cbe11f added .gitignore into main
* 304cb7d initialized empty repo
```

Настраивает формат вывода коммитов (флаг **--pretty**):

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git log --all --pretty=format:"%h | %an | %ar | %s"
3c10766 | Angelina Malei | 73 minutes ago | created IGI_LR5 branch
eb7d510 | Angelina Malei | 75 minutes ago | created IGI_LR4 branch
11d6ac0 | Angelina Malei | 77 minutes ago | created IGI_LR3 branch
17be4b6 | Angelina Malei | 80 minutes ago | created IGI_LR2 branch
70691f8 | Angelina Malei | 84 minutes ago | created IGI_LR1 branch
971c9b5 | Angelina Malei | 88 minutes ago | created STRWEB_LR4 branch
b5ab026 | Angelina Malei | 2 hours ago | created SRTWEB_LR3 branch
1de7e07 | Angelina Malei | 2 hours ago | created STRWEB_LR2 branch
a7d2444 | Angelina Malei | 2 hours ago | created STRWEB_LR1 branch
e76c10b | Angelina Malei | 2 hours ago | created STRWEB branch
1c84e30 | Angelina Malei | 2 hours ago | created IGI branch
77deff2 | Angelina Malei | 2 hours ago | committed all .txt files
4cbe11f | Angelina Malei | 2 hours ago | added .gitignore into main
304cb7d | Angelina Malei | 2 hours ago | initialized empty repo
```

Показывает даты относительно текущего момента (флаг **--relative-date**):

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git log --all --relative-date
commit 3c10766f5719950660f013558ee5436c97f52b6d (IGI_LR5)
Author: Angelina Malei <maley.gelya@mail.ru>
Date:   75 minutes ago

    created IGI_LR5 branch

commit eb7d5106cb4d9ae26d4671f01877d50ff5255232 (IGI_LR4)
Author: Angelina Malei <maley.gelya@mail.ru>
Date:   77 minutes ago

    created IGI_LR4 branch
```

i) Найти разницу в изменениях для каждого коммита

История изменений для каждого коммита (**-p**):

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git log --all -p
commit 3c10766f5719950660f013558ee5436c97f52b6d (IGI_LR5)
Author: Angelina Malei <maley.gelya@mail.ru>
Date:   Sun Feb 22 19:49:33 2026 +0300

    created IGI_LR5 branch

diff --git a/IGI/IGITest.txt b/IGI/IGITest.txt
deleted file mode 100644
index e69de29..0000000
diff --git a/IGI/LR1/LR1Test.txt b/IGI/LR1/LR1Test.txt
deleted file mode 100644
index e69de29..0000000
diff --git a/IGI/LR1/bar/baz/readme.txt b/IGI/LR1/bar/baz/readme.txt
deleted file mode 100644
index 1e6fd03..0000000
--- a/IGI/LR1/bar/baz/readme.txt
+++ /dev/null
@@ -1 +0,0 @@
-444
```

Изменения конкретного коммита (**git show**):

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git show 4cbe11f
commit 4cbe11f6e8437f6149e8dadcb0f78a5119dcd61d
Author: Angelina Malei <maley.gelya@mail.ru>
Date:   Sun Feb 22 18:52:15 2026 +0300

    added .gitignore into main

diff --git a/.gitignore b/.gitignore
new file mode 100644
index 0000000..3fdb35e
--- /dev/null
+++ b/.gitignore
@@ -0,0 +1 @@
+IGI/LR1/geometric_lib/
```

Незакоммиченные изменения или разница между объектами (**git diff**):

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git diff 77deff2 4cbe11f
diff --git a/IGI/IGITest.txt b/IGI/IGITest.txt
deleted file mode 100644
index e69de29..0000000
diff --git a/IGI/LR1/LR1Test.txt b/IGI/LR1/LR1Test.txt
deleted file mode 100644
index e69de29..0000000
diff --git a/IGI/LR1/bar/baz/readme.txt b/IGI/LR1/bar/baz/readme.txt
deleted file mode 100644
index 1e6fd03..0000000
--- a/IGI/LR1/bar/baz/readme.txt
+++ /dev/null
@@ -1 +0,0 @@
-444
```

j) Создать новый файл test.txt и вставить в него строчку 'номер варианта'. Затем сделать коммит и отменить его

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ echo "номер варианта 19" > test.txt

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git status
On branch main
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        test.txt

nothing added to commit but untracked files present (use "git add" to track)

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git add test.txt
warning: in the working copy of 'test.txt', LF will be replaced by CRLF the n

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git status
On branch main
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   test.txt


Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git commit -m "add new test.txt"
[main 386c6e8] add new test.txt
 1 file changed, 1 insertion(+)
 create mode 100644 test.txt

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git reset --soft 386c6e8

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git status
On branch main
nothing to commit, working tree clean

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ ls
453503_MALEY_19Test.txt  IGI/  STRWEB/  test.txt
```

k) Проиндексировать файл в своем Git-репозитории, затем отменить его индексацию

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ echo "test index" >> test.txt

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git status
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   test.txt

no changes added to commit (use "git add" and/or "git commit -a")

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git add test.txt
warning: in the working copy of 'test.txt', LF will be replaced by CRLF

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git status
On branch main
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   test.txt


Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git reset test.txt
Unstaged changes after reset:
M       test.txt

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git status
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   test.txt

no changes added to commit (use "git add" and/or "git commit -a")
```

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git add test.txt
warning: in the working copy of 'test.txt', LF will be replaced by CRLF

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git status
On branch main
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   test.txt


Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git restore --staged test.txt

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git status
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   test.txt

no changes added to commit (use "git add" and/or "git commit -a")
```

l) Внести изменения в файл в своем Git-репозитории и затем вернуть его в исходное состояние.

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git status
On branch main
nothing to commit, working tree clean

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ cat test.txt
номер варианта 19

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ echo "some changes" >> test.txt

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ cat test.txt
номер варианта 19
some changes

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git status
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   test.txt

no changes added to commit (use "git add" and/or "git commit -a")
```

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git restore test.txt

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git status
On branch main
nothing to commit, working tree clean

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ cat test.txt
номер варианта 19
```

<u>5 Выполнить следующие действия с удаленным репозиторием:</u>

a) Создать удаленный репозиторий (на github.com)

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git remote add origin https://github.com/gelya-maley/453503_MALEY_19.git

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git branch -M main

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git push -u origin main
info: please complete authentication in your browser...
Enumerating objects: 27, done.
Counting objects: 100% (27/27), done.
Delta compression using up to 20 threads
Compressing objects: 100% (14/14), done.
Writing objects: 100% (27/27), 2.15 KiB | 146.00 KiB/s, done.
Total 27 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), done.
To https://github.com/gelya-maley/453503_MALEY_19.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

b) Просмотреть информацию о нем, внести изменения и опять просмотреть (remote, fetch).

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git remote show origin
* remote origin
  Fetch URL: https://github.com/gelya-maley/453503_MALEY_19.git
  Push  URL: https://github.com/gelya-maley/453503_MALEY_19.git
  HEAD branch: main
  Remote branches:
    IGI        tracked
    IGI_LR1    tracked
    IGI_LR2    tracked
    IGI_LR3    tracked
    IGI_LR4    tracked
    IGI_LR5    tracked
    STRWEB     tracked
    STRWEB_LR1 tracked
    STRWEB_LR2 tracked
    STRWEB_LR3 tracked
    STRWEB_LR4 tracked
    main       tracked
  Local branch configured for 'git pull':
    main merges with remote main
  Local refs configured for 'git push':
    IGI        pushes to IGI        (up to date)
    IGI_LR1    pushes to IGI_LR1    (up to date)
    IGI_LR2    pushes to IGI_LR2    (up to date)
    IGI_LR3    pushes to IGI_LR3    (up to date)
    IGI_LR4    pushes to IGI_LR4    (up to date)
    IGI_LR5    pushes to IGI_LR5    (up to date)
    STRWEB     pushes to STRWEB     (up to date)
    STRWEB_LR1 pushes to STRWEB_LR1 (up to date)
    STRWEB_LR2 pushes to STRWEB_LR2 (up to date)
    STRWEB_LR3 pushes to STRWEB_LR3 (up to date)
    STRWEB_LR4 pushes to STRWEB_LR4 (up to date)
    main       pushes to main       (up to date)
```

c) Изменить несколько файлов в своем проекте и перенести их в удаленный репозиторий

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ echo "Внесены изменения для удаленного репозитория" >> 453503_MALEY_19Test.txt

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ cat 453503_MALEY_19Test.txt
Инициализирован пустой репозиторий
Внесены изменения для удаленного репозитория

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   453503_MALEY_19Test.txt

no changes added to commit (use "git add" and/or "git commit -a")
```

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git add 453503_MALEY_19Test.txt
warning: in the working copy of '453503_MALEY_19Test.txt', LF will be

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   453503_MALEY_19Test.txt


Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git commit -m "modified 453503_MALEY_19Test.txt"
[main 8de2313] modified 453503_MALEY_19Test.txt
 1 file changed, 1 insertion(+)

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 20 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 377 bytes | 377.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/gelya-maley/453503_MALEY_19.git
   386c6e8..8de2313  main -> main
```

d) Переименовать свой удаленный репозиторий

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git remote -v
origin  https://github.com/gelya-maley/453503_MALEY_19.git (fetch)
origin  https://github.com/gelya-maley/453503_MALEY_19.git (push)

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git remote rename original renamed
error: No such remote: 'original'

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git remote rename origin renamed

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git remote -v
renamed https://github.com/gelya-maley/453503_MALEY_19.git (fetch)
renamed https://github.com/gelya-maley/453503_MALEY_19.git (push)
```

e) Получить данные из удаленного репозитория

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git pull renamed main
From https://github.com/gelya-maley/453503_MALEY_19
 * branch            main       -> FETCH_HEAD
Already up to date.
```

f) Просмотреть историю изменений

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git log --oneline --all -5
8de2313 (HEAD -> main, renamed/main) modified 453503_MALEY_19Test.txt
386c6e8 add new test.txt
522a2c4 Revert "add test.txt with variant number"
0092a82 add test.txt with variant number
3c10766 (renamed/IGI_LR5, IGI_LR5) created IGI_LR5 branch
```

g) Создать дополнительную ветку 1 в удаленном репозитории, перенести в нее два файла из другой ветки, удалить один из файлов

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ echo "from root dir to branch1" > proverka.txt

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ echo "from root dir to branch1" > test.txt
```

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git status
On branch main
Your branch is up to date with 'renamed/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   test.txt

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        proverka.txt

no changes added to commit (use "git add" and/or "git commit -a")

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git add .
warning: in the working copy of 'test.txt', LF will be replaced by CRLF 
warning: in the working copy of 'proverka.txt', LF will be replaced by CR
t

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git commit -m "added 2 .txt files"
[main 2b8da7c] added 2 .txt files
 2 files changed, 2 insertions(+), 1 deletion(-)
 create mode 100644 proverka.txt

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git status
On branch main
Your branch is ahead of 'renamed/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
```

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git checkout -b branch1
Switched to a new branch 'branch1'

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (branch1)
$ git status
On branch branch1
nothing to commit, working tree clean

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (branch1)
$ git checkout main -- proverka.txt test.txt
```

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (branch1)
$ git rm proverka.txt
rm 'proverka.txt'

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (branch1)
$ git status
On branch branch1
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        deleted:    proverka.txt


Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (branch1)
$ git commit -m "deleted 1 file after 2 files have been added to branch"
[branch1 5a644f0] deleted 1 file after 2 files have been added to branch
 1 file changed, 1 deletion(-)
 delete mode 100644 proverka.txt

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (branch1)
$ git status
On branch branch1
nothing to commit, working tree clean
```

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (branch1)
$ git push -u renamed branch1
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 20 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 517 bytes | 517.00 KiB/s, done.
Total 5 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 1 local object.
remote:
remote: Create a pull request for 'branch1' on GitHub by visiting:
remote:      https://github.com/gelya-maley/453503_MALEY_19/pull/new/branch1
remote:
To https://github.com/gelya-maley/453503_MALEY_19.git
 * [new branch]      branch1 -> branch1
branch 'branch1' set up to track 'renamed/branch1'.
```

h) Создать дополнительную ветку 2 в удаленном репозитории, перенести в нее два файла из своего проекта, слить изменения в основную ветку, удалить дополнительную ветку 2

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git checkout -b branch2
Switched to a new branch 'branch2'
```

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (branch2)
$ git checkout main -- 453503_MALEY_19Test.txt test.txt
```

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (branch2)
$ echo "add 2 files to branch2" >> 453503_MALEY_19Test.txt

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (branch2)
$ echo "add 2 files to branch2" >> test.txt
```

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (branch2)
$ git status
On branch branch2
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   453503_MALEY_19Test.txt
        modified:   test.txt

no changes added to commit (use "git add" and/or "git commit -a")
```

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (branch2)
$ git add 453503_MALEY_19Test.txt test.txt
warning: in the working copy of '453503_MALEY_19Test.txt', LF will be re
t touches it
warning: in the working copy of 'test.txt', LF will be replaced by CRLF

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (branch2)
$ git status
On branch branch2
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   453503_MALEY_19Test.txt
        modified:   test.txt


Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (branch2)
$ git commit -m "added 2 files to branch2"
[branch2 7117f86] added 2 files to branch2
 2 files changed, 2 insertions(+)
```

```
Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (branch2)
$ git push renamed branch2
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 20 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 392 bytes | 392.00 KiB/s, done.
Total 4 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
remote:
remote: Create a pull request for 'branch2' on GitHub by visiting:
remote:      https://github.com/gelya-maley/453503_MALEY_19/pull/new/branch2
remote:
To https://github.com/gelya-maley/453503_MALEY_19.git
 * [new branch]      branch2 -> branch2

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (branch2)
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'renamed/main'.

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git merge branch2
Updating 2b8da7c..7117f86
Fast-forward
 453503_MALEY_19Test.txt | 1 +
 test.txt                | 1 +
 2 files changed, 2 insertions(+)

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git push renamed main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/gelya-maley/453503_MALEY_19.git
   2b8da7c..7117f86  main -> main

Angelina@angelpc MINGW64 ~/Desktop/4sem/ИГИ/453503_MALEY_19 (main)
$ git branch -D branch2
Deleted branch branch2 (was 7117f86).
```

6 https://gelya-maley.github.io/IGI-webpage/